

From Functions to Hessians

This guide is designed for students beginning their journey into the mathematics of machine learning. It starts from the very basics—what a function is—and builds up to gradients and Hessians, with an emphasis on *rules, intuition, and concrete examples*, especially for *linear and quadratic functions*. It also includes the *chain rule (single and multivariable)* and the *product rule for gradients*.

1 What is a Function?

A **function** is a rule that maps an input to a single output.

- **Scalar** $\rightarrow$ **scalar**: $f : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto f(x)$. Example: $f(x) = x^2$.
- **Vector** $\rightarrow$ **scalar** (common for ML losses): $f : \mathbb{R}^n \rightarrow \mathbb{R}, \mathbf{x} \mapsto f(\mathbf{x})$. Example: $f(\mathbf{x}) = \|\mathbf{x}\|^2$.
- **Vector** $\rightarrow$ **vector**: $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$. Example: $g(\mathbf{x}) = A\mathbf{x}$ (a linear layer).

Notation. Scalars: x, a, b, c . Vectors: $\mathbf{x} \in \mathbb{R}^n$ (column). Matrices: $A \in \mathbb{R}^{m \times n}$ (we use A for matrices consistently). Transpose: $\mathbf{x}^\top$. Euclidean norm: $\|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2}$. Identity matrix: I (dimension clear from context).

2 Derivative (Single-Variable)

The **derivative** measures instantaneous rate of change:

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}.$$

Intuition: the slope of f at x . **Examples:** $\frac{d}{dx}x^k = kx^{k-1}$, $\frac{d}{dx}e^x = e^x$, $\frac{d}{dx}\ln x = 1/x$ (for $x > 0$).

Sigmoid: $\sigma(t) = \frac{1}{1+e^{-t}}$, $\sigma'(t) = \sigma(t)(1 - \sigma(t))$. ReLU: $\text{ReLU}(t) = \max\{0, t\}$; derivative = 0 if $t < 0$, = 1 if $t > 0$ (use a subgradient at $t = 0$).

3 Gradient (Multivariable Derivative)

For $f : \mathbb{R}^n \rightarrow \mathbb{R}$, the **gradient** is

$$\nabla f(\mathbf{x}) = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \vdots \\ \frac{\partial f}{\partial x_n} \end{bmatrix}.$$

Intuition: direction of *steepest ascent* ($-\nabla f$ for descent). **Gradient descent:** $\mathbf{x}_{k+1} = \mathbf{x}_k - \eta \nabla f(\mathbf{x}_k)$. **Directional derivative** in unit direction $\mathbf{v}$: $D_{\mathbf{v}}f(\mathbf{x}) = \nabla f(\mathbf{x})^\top \mathbf{v}$.

Core Gradient Rules

Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ and $a, b \in \mathbb{R}$.

Linearity	$\nabla(af + bg) = a\nabla f + b\nabla g$
Product	$\nabla(fg) = f\nabla g + g\nabla f$
Quotient	$\nabla\left(\frac{f}{g}\right) = \frac{g\nabla f - f\nabla g}{g^2} \quad (g \neq 0)$

Chain Rule

Single-variable: If $h(x) = f(g(x))$, then $h'(x) = f'(g(x))g'(x)$. Example: $h(x) = \sin(x^2) \Rightarrow h'(x) = \cos(x^2) \cdot 2x$.

Multivariable (vector form): Let $h(\mathbf{x}) = f(g(\mathbf{x}))$ with $g : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $f : \mathbb{R}^m \rightarrow \mathbb{R}$. Then

$$\nabla_{\mathbf{x}} h(\mathbf{x}) = J_g(\mathbf{x})^\top \nabla_{\mathbf{u}} f(\mathbf{u}) \Big|_{\mathbf{u}=g(\mathbf{x})},$$

where $J_g(\mathbf{x}) \in \mathbb{R}^{m \times n}$ is the Jacobian. This identity underlies *backpropagation*.

Handy Gradients (Memorize These!)

$f(\mathbf{x})$	$\nabla f(\mathbf{x})$
$\frac{1}{2}\ \mathbf{x}\ _2^2$	$\mathbf{x}$
$\mathbf{a}^\top \mathbf{x}$	$\mathbf{a}$
$\frac{1}{2}\ A\mathbf{x} - \mathbf{b}\ _2^2$	$A^\top(A\mathbf{x} - \mathbf{b})$
$\log(1 + e^{\mathbf{w}^\top \mathbf{x}})$	$\sigma(\mathbf{w}^\top \mathbf{x}) \mathbf{w}$

Derivations for the Table (worked out)

We use bold lower-case for vectors, A for matrices, and standard chain/differential rules.

(1) $f(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|_2^2$. Note $\|\mathbf{x}\|_2^2 = \mathbf{x}^\top \mathbf{x}$.

$$df = \frac{1}{2} d(\mathbf{x}^\top \mathbf{x}) = \frac{1}{2} ((d\mathbf{x})^\top \mathbf{x} + \mathbf{x}^\top d\mathbf{x}) = \mathbf{x}^\top d\mathbf{x}.$$

Since $df = \nabla f(\mathbf{x})^\top d\mathbf{x}$ for all $d\mathbf{x}$, we obtain $\boxed{\nabla f(\mathbf{x}) = \mathbf{x}}$ and $\nabla^2 f = I$.

(2) $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x}$. $df = \mathbf{a}^\top d\mathbf{x} \Rightarrow \boxed{\nabla f(\mathbf{x}) = \mathbf{a}}$, and $\nabla^2 f = \mathbf{0}$.

(3) $f(\mathbf{x}) = \frac{1}{2}\|A\mathbf{x} - \mathbf{b}\|_2^2$. Let $\mathbf{u} = A\mathbf{x} - \mathbf{b}$. Then $df = \mathbf{u}^\top d\mathbf{u}$ and $d\mathbf{u} = A d\mathbf{x}$. Thus

$$df = \mathbf{u}^\top (A d\mathbf{x}) = (A^\top \mathbf{u})^\top d\mathbf{x} \Rightarrow \boxed{\nabla f(\mathbf{x}) = A^\top(A\mathbf{x} - \mathbf{b})},$$

and $\nabla^2 f = A^\top A \succeq 0$.

(4) $f(\mathbf{x}) = \log(1 + e^{\mathbf{w}^\top \mathbf{x}})$. Let $z = \mathbf{w}^\top \mathbf{x}$, $f = \phi(z)$ with $\phi(z) = \log(1 + e^z)$. Then $\phi'(z) = \frac{e^z}{1+e^z} = \sigma(z)$, so

$$\nabla_{\mathbf{x}} f = \phi'(z) \nabla_{\mathbf{x}} z = \sigma(z) \mathbf{w} = \boxed{\sigma(\mathbf{w}^\top \mathbf{x}) \mathbf{w}}.$$

(5) **Bonus:** $f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top A \mathbf{x}$ (general quadratic).

$$\begin{aligned} df &= \frac{1}{2} d(\mathbf{x}^\top A \mathbf{x}) = \frac{1}{2} \left((d\mathbf{x})^\top A \mathbf{x} + \mathbf{x}^\top A d\mathbf{x} \right) \\ &= \frac{1}{2} \left((A\mathbf{x})^\top d\mathbf{x} + (A^\top \mathbf{x})^\top d\mathbf{x} \right) = \frac{1}{2} (A + A^\top) \mathbf{x}^\top d\mathbf{x}. \end{aligned}$$

Hence

$$\boxed{\nabla f(\mathbf{x}) = \frac{1}{2} (A + A^\top) \mathbf{x}}, \quad \boxed{\nabla^2 f = \frac{1}{2} (A + A^\top)}.$$

If A is symmetric, then $\nabla f = A\mathbf{x}$ and $\nabla^2 f = A$.

4 Hessian (Matrix of Second Derivatives)

The **Hessian** $\nabla^2 f(\mathbf{x})$ is the $n \times n$ matrix $[\nabla^2 f(\mathbf{x})]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$. It is symmetric if $f \in C^2$. **Curvature:** describes how the gradient changes. **Newton's method:** $\mathbf{x}_{k+1} = \mathbf{x}_k - [\nabla^2 f(\mathbf{x}_k)]^{-1} \nabla f(\mathbf{x}_k)$. **Convexity:** f is convex if $\nabla^2 f(\mathbf{x}) \succeq 0$ for all $\mathbf{x}$. At a stationary point $\nabla f = 0$: positive definite Hessian $\Rightarrow$ strict local minimum; negative definite $\Rightarrow$ strict local maximum; indefinite $\Rightarrow$ saddle.

5 Linear & Quadratic Functions (Analyzed)

Linear Function

$f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$. Then $\nabla f = \mathbf{a}$ (constant) and $\nabla^2 f = \mathbf{0}$ (no curvature). Geometrically a flat plane; without constraints, no finite minimizer.

Quadratic Function (General Form)

$$f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top A \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c.$$

Gradient: $\nabla f(\mathbf{x}) = \frac{1}{2} (A + A^\top) \mathbf{x} + \mathbf{b}$. If A is symmetric, $\nabla f = A\mathbf{x} + \mathbf{b}$. Hessian: $\nabla^2 f = \frac{1}{2} (A + A^\top)$ (constant). Convexity: f is convex iff $A_{\text{sym}} = \frac{1}{2} (A + A^\top) \succeq 0$. If $A \succ 0$, the unique minimizer satisfies $\nabla f(\mathbf{x}^*) = 0 \Rightarrow A\mathbf{x}^* + \mathbf{b} = 0 \Rightarrow \mathbf{x}^* = -A^{-1}\mathbf{b}$.

Least Squares (Quadratic Special Case)

$$f(\mathbf{x}) = \frac{1}{2} \|A\mathbf{x} - \mathbf{b}\|_2^2 = \frac{1}{2} \mathbf{x}^\top (A^\top A) \mathbf{x} - \mathbf{b}^\top A\mathbf{x} + \frac{1}{2} \|\mathbf{b}\|^2.$$

Gradient: $\nabla f = A^\top (A\mathbf{x} - \mathbf{b})$. Hessian: $\nabla^2 f = A^\top A \succeq 0$. If $A^\top A$ is invertible, the minimizer is $\mathbf{x}^* = (A^\top A)^{-1} A^\top \mathbf{b}$.

6 Why This Matters in Machine Learning

Training typically minimizes a loss $f(\mathbf{w})$. **Gradient Descent:** $\mathbf{w}_{k+1} = \mathbf{w}_k - \eta \nabla f(\mathbf{w}_k)$. **Newton's Method:** $\mathbf{w}_{k+1} = \mathbf{w}_k - [\nabla^2 f(\mathbf{w}_k)]^{-1} \nabla f(\mathbf{w}_k)$. Linear/quadratic forms appear in linear/logistic regression, ridge/L2 regularization, Gaussian models, and as local quadratic approximations via Taylor expansion.

7 Quick Practice (With Answers)

1. $f(x) = 3x^3 - 2x$. Find $f'(x)$. **Ans:** $9x^2 - 2$.
2. $f(\mathbf{x}) = \frac{1}{2}\|\mathbf{x}\|_2^2 + \mathbf{a}^\top \mathbf{x}$. Compute ∇f and $\nabla^2 f$. **Ans:** $\nabla f = \mathbf{x} + \mathbf{a}$, $\nabla^2 f = I$.
3. $f(\mathbf{x}) = \frac{1}{2}\|A\mathbf{x} - \mathbf{b}\|^2$. Show $\nabla f = A^\top(A\mathbf{x} - \mathbf{b})$ via the chain rule. **Ans:** Let $\mathbf{u} = A\mathbf{x} - \mathbf{b}$, $h(\mathbf{u}) = \frac{1}{2}\|\mathbf{u}\|^2$. Then $\nabla h = \mathbf{u}$, $J_g = A$. Hence $\nabla f = A^\top \mathbf{u} = A^\top(A\mathbf{x} - \mathbf{b})$.
4. $f(\mathbf{x}) = (\mathbf{a}^\top \mathbf{x})(\mathbf{b}^\top \mathbf{x})$. Find ∇f . **Ans:** Using the product rule, $\nabla f = (\mathbf{b}^\top \mathbf{x}) \mathbf{a} + (\mathbf{a}^\top \mathbf{x}) \mathbf{b}$.

8 Advanced Example: Logistic Loss

Let $f(\mathbf{w}) = \log(1 + \exp(-y \mathbf{w}^\top \mathbf{x}))$ for binary classification. Set $z = y \mathbf{w}^\top \mathbf{x}$, so $f = \log(1 + e^{-z})$. Then $\frac{df}{dz} = -\sigma(-z) = \sigma(z) - 1$, and

$$\nabla f = \frac{df}{dz} \cdot \frac{\partial z}{\partial \mathbf{w}} = (\sigma(z) - 1) \cdot (y \mathbf{x}) = -y \sigma(-y \mathbf{w}^\top \mathbf{x}) \mathbf{x},$$

which is the standard gradient used in logistic regression training.

Multivariate Taylor Expansion (Local Quadratic Approximation)

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be C^2 . For a small step $\Delta \mathbf{x}$,

$$f(\mathbf{x} + \Delta \mathbf{x}) \approx f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^\top [\nabla^2 f(\mathbf{x})] \Delta \mathbf{x}.$$

- **Linear** $f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} + b$: $\nabla f = \mathbf{a}$, $\nabla^2 f = \mathbf{0}$. Taylor is exact at first order: $f(\mathbf{x} + \Delta \mathbf{x}) = f(\mathbf{x}) + \mathbf{a}^\top \Delta \mathbf{x}$.
- **Quadratic** $f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^\top A \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c$: $\nabla f = \frac{1}{2}(A + A^\top) \mathbf{x} + \mathbf{b}$, $\nabla^2 f = \frac{1}{2}(A + A^\top)$. The quadratic Taylor model is exact:

$$f(\mathbf{x} + \Delta \mathbf{x}) = f(\mathbf{x}) + \nabla f(\mathbf{x})^\top \Delta \mathbf{x} + \frac{1}{2} \Delta \mathbf{x}^\top [\frac{1}{2}(A + A^\top)] \Delta \mathbf{x}.$$

If A is symmetric, this simplifies with $\nabla f = A\mathbf{x} + \mathbf{b}$ and $\nabla^2 f = A$.

- **Least squares** $f(\mathbf{x}) = \frac{1}{2}\|A\mathbf{x} - \mathbf{b}\|_2^2$: $\nabla f = A^\top(A\mathbf{x} - \mathbf{b})$, $\nabla^2 f = A^\top A \succeq 0$. A Newton step $\Delta \mathbf{x}$ solves $(A^\top A)\Delta \mathbf{x} = -A^\top(A\mathbf{x} - \mathbf{b})$, so $\Delta \mathbf{x} = -(A^\top A)^{-1}A^\top(A\mathbf{x} - \mathbf{b})$ whenever $A^\top A$ is invertible.